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Roll-no: 24K-0737

Question #1:-

Relations on $\{a, b, c, d\}$

$$R = \{(a, a), (a, c), (a, d), (b, a), (b, b), (b, c), (b, d), (c, b), (c, c), (d, b), (d, d)\}$$

- (a) Reflexive: Yes. All (x, x) appears. $(a, a), (b, b), (c, c), (d, d)$
- (b) Symmetric: No. $(a, c) \in R$ but $(c, a) \notin R$
- (c) Anti-symmetric: No. $(b, c) \in R$ and $(c, b) \in R$. $b \neq c$
- (d) Transitive: No. $(a, c) \in R$ and $(c, b) \in R$ but $(a, b) \notin R$
- (e) Irreflexive: No. b/c it contains (x, x) pairs
- (f) Asymmetric: No. (b, c) and (c, b)

Question #2:-

$$A = \{0, 1, 2, 3, 4\} \text{ to } B = \{0, 1, 2, 3\}$$

(a) $a = b$
 $\{ (0, 0), (1, 1), (2, 2), (3, 3) \}$ Ans.

(b) $a + b = 4$
 $\{ (1, 3), (2, 2), (3, 1), (4, 0) \}$ Ans.

(c) $a > b$
 $\{ (1, 0), (2, 0), (2, 1), (3, 0), (3, 1), (3, 2), (4, 0), (4, 1), (4, 2), (4, 3) \}$

Ans.

(d) $a/b :-$

Soln.

$\{(1,0), (1,1), (1,2), (1,3), (2,0), (2,2), (3,0), (3,3), (4,0)\} \cap M_1$

(e) $GCD(a,b) = 1$

Soln.

$\{(0,1), (1,0), (1,1), (1,2), (1,3), (2,1), (2,3), (3,1), (3,2), (4,1), (4,3)\} \cap M_2$

(f) $LCM(a,b) = 2$

Soln.

$\{(1,2), (2,1), (2,2)\} \cap M_2$

(b) Question #3:-

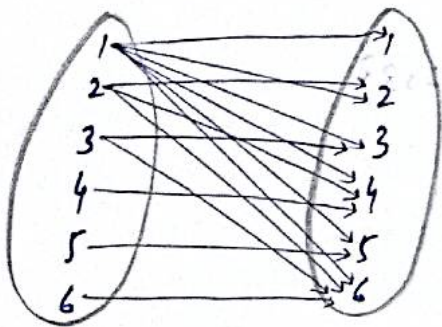
$R = \{(a,b) \mid a/b\}$ on $\{1,2,3,4,5,6\}$ ($1 \leq a, b \leq 6$)

Soln.

(i) $\{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,2), (2,4), (2,6), (3,3), (3,6), (4,4), (5,5), (6,6)\} \cap M_1$

(ii) (a)

(i) Directed graph:-



(b) $\Rightarrow$ It is not equivalence (not symmetric, in general).

$\rightarrow$ It is a partial order relation if divisibility is reflexive, anti-symmetric.

(c) matrix-form:-

	1	2	3	4	5	6
1	1	1	1	1	1	1
2	0	1	0	1	0	1
3	0	0	1	0	0	1
4	0	0	0	1	0	0
5	0	0	0	0	1	0
6	0	0	0	0	0	1

Yash-Raj

Roll-no:- 24K-0737

Question #4:-

(a) $R = \{(2,2), (2,3), (2,4), (3,2), (3,3), (3,4)\}$

- Reflexive = No. $(1,1), (4,4)$ missing.
- Anti-symmetric = No. $(2,3) \in R$ & $(3,2) \in R$ $2 \neq 3$
- Symmetric = Yes. $(2,4), (4,2)$
- Transitive = Yes

Ans.

(b) $R = \{(1,1), (1,2), (2,1), (2,2), (3,3), (4,4)\}$

Soln:-

- Reflexive = Yes.
- Anti-symmetric = Yes
- Symmetric = No
- Transitive = Yes

Ans.

(c) $R = \{(2,4), (4,2)\}$

Soln:-

- Reflexive = No
- Anti-symmetric = No
- Symmetric = Yes
- Transitive = No

Ans.

Question #6:-

(a) ~~a is taller than b~~

(d) $R = \{(1,2), (2,3), (3,4)\}$
Soln

- Reflexive: No
- Symmetric: No
- Anti-symmetric: Yes
- Transitive: No

(e) $R = \{(1,1), (2,2), (3,3), (4,4)\}$ Ans.

- Soln
- Reflexive: Yes
 - Symmetric: Yes
 - Anti-symmetric: Yes
 - Transitive: Yes

(f) $R = \{(1,3), (1,4), (2,3), (2,4), (3,1), (3,4)\}$

- Soln
- Reflexive: No
 - Symmetric: No
 - Anti-symmetric: No
 - Transitive: No

Question #5:-

(a) "a is taller than b"

- Reflexive: No
- Irreflexive: Yes
- Symmetric: No
- Anti-symmetric: Yes
- Asymmetric: Yes
- Transitive: Yes

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I-no:- 24k-01
Question #1:-

me:- Yash-Raj

Discrete Assignment 02

(3)

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Question #5 (b):- a and b were born on same day.

son:-

Reflexive = Yes

Symmetric = Yes

Transitive = Yes

Anti-symmetric = No

Asymmetric = No

(c) a has the same first name as b:-

Reflexive = Yes

Symmetric = Yes

Transitive = Yes

Anti-symmetric = No

Asymmetric = No

Irreflexive = No

(d) a and b have a common grandparent:-

son:-

Reflexive = Yes

Symmetric = Yes

Transitive = No

Anti-symmetric = No

Question #6:

(a) Both symmetric and anti-symmetric:

Ans: A relation that is both must contain only (x, x)

Ex: identity relation

$R = \{(n, n) | n \in \mathbb{N}\}$. It is symmetric, Anti-symmetric, transitive.

• Empty set is not reflexive

(b) Not symmetric and not anti-symmetric:

Ans: One construction: include a mutual pair for distinct elements (to break anti-symmetry) and also include a pair whose reverse is missing (to break symmetry).

$R = \{(0, 1), (1, 0), (2, 3)\}$

• $(0, 1)$ and $(1, 0)$ together make anti-symmetric.

• $(2, 3)$ with $(3, 2)$ makes symmetry fail.

(c) Both symmetric and Transitive:

Ans: Any equivalence relation works. For example: $R = \{(x, y) | n\}$
so, the identity is symmetric and transitive.

(d) Neither symmetric nor Transitive:

Ans: $\{(0, 1), (1, 2)\}$. Not symmetric (no-reverse) and not transitive because $(0, 1)$ and $(1, 2)$ are in R but $(0, 2) \notin R$.

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$$A = \{1, 2, 3\}$$

Question #7:

$$R_1 = \{(a, b) \mid a > b\} \Rightarrow \{(2, 1), (3, 1), (3, 2)\}$$

$$R_2 = \{(a, b) \mid a \geq b\} \Rightarrow \{(1, 1), (2, 1), (2, 2), (3, 1), (3, 2), (3, 3)\}$$

$$R_3 = \{(a, b) \mid a < b\} \Rightarrow \{(1, 2), (1, 3), (2, 3)\}$$

$$R_4 = \{(a, b) \mid a \leq b\} \Rightarrow \{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$$

$$R_5 = \{(a, b) \mid a = b\} \Rightarrow \{(1, 1), (2, 2), (3, 3)\}$$

$$R_6 = \{(a, b) \mid a \neq b\} \Rightarrow \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\}$$

$$(a) R_2 \cup R_4$$

SDA
(AXA)
= all ordered pairs
Ans

$$(b) R_3 \cup R_1$$

SDA
R6 Ans

$$(c) R_3 \cap R_6$$

SDA
R3 Ans

$$(d) R_4 \cap R_6$$

SDA
R3 Ans

$$(e) R_3 - R_6$$

SDA
R6 Ans

$$(f) R_6 - R_3$$

SDA
 ~~$\{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$~~
R1 Ans

$$(g) R_2 \oplus R_6$$

SDA
 $\{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$

$$(h) R_3 \oplus R_5$$

SDA
 $\{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$

$$(i) R_2 \circ R_1$$

SDA
R1

$$(j) R_6 \circ R_6$$

SDA
AXA
(all 9
ordered
pairs)

Ans

Question #8

$A = \{1, 2, 3\}$ with a relation (Transitive)

(i) $\{(1,1), (1,2), (1,3)\}$

Soln.

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ Ans.}$$

(iv) $\{(1,3), (3,1)\}$

Soln.

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \text{ Ans.}$$

(ii) $R = \{(1,2), (2,1), (2,2), (3,3)\}$

Soln.

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ Ans.}$$

(iii) $R = \{(1,1), (1,2), (1,3), (2,2), (2,3), (3,3)\}$

Soln.

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \text{ Ans.}$$

part (b):-

(i) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

Soln.

$$= \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$$

 Ans.

(iii) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

Soln.

$$= \{(1,2), (2,2), (3,2)\}$$

 Ans.

(iii) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Soln.

$$= \{(1,1), (1,2), (1,3), (2,1), (2,3), (3,1), (3,2), (3,3)\}$$

 Ans.

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Question 9:-

(a)

Reason:- Reflexive $\mathcal{L}(a) \leq \mathcal{L}(a)$

Symmetric $\mathcal{L}(a) \leq \mathcal{L}(b)$ then $\mathcal{L}(b) \leq \mathcal{L}(a)$

Transitive $\mathcal{L}(a) \leq \mathcal{L}(b)$ and $\mathcal{L}(b) \leq \mathcal{L}(c)$ then $\mathcal{L}(a) \leq \mathcal{L}(c)$ Ans.

(b) Reflexive $a \equiv a$

Symmetric if $a \equiv b$ then $b \equiv a$

Transitive $a \equiv b$ and $b \equiv c$ then $a \equiv c$ Ans.

Question 10 (Part a):-

(a) $2^n - 1$

Soln.

$$2^1 - 1 = 1$$

$$2^2 - 1 = 3$$

$$2^3 - 1 = 7$$

$$2^4 - 1 = 15$$

$\therefore 1, 3, 7, 15$ Ans.

(c) $\frac{(-1)^n}{n^2}$

Soln.

$$\frac{(-1)^1}{(1)^2} = -1$$

$$\frac{(-1)^2}{(2)^2} = \frac{1}{4} \quad \therefore -1, \frac{1}{4}, -\frac{1}{9}, \frac{1}{16}, -\frac{1}{25}$$

$$\frac{(-1)^3}{(3)^2} = -\frac{1}{9}$$

$$\frac{(-1)^4}{(4)^2} = \frac{1}{16}$$

$$\frac{(-1)^5}{(5)^2} = -\frac{1}{25}$$

(b) $10 - \frac{3n}{2}$

Soln.

$$10 - \frac{3(1)}{2} = 17/2$$

$$10 - \frac{3(2)}{2} = 7$$

$$10 - \frac{3(3)}{2} = 11/2$$

$$10 - \frac{3(4)}{2} = 4$$

$$10 - \frac{3(5)}{2} = 5/2$$

$\therefore 17/2, 7, 11/2, 4, 5/2$ Ans.

(iv) $\frac{3n+4}{2n-1}$

Soln.

$$\frac{3(1)+4}{2(1)-1} = 7$$

$$\frac{3(2)+4}{2(2)-1} = 10/3$$

$$\frac{3(3)+4}{2(3)-1} = 13/5$$

$$\frac{3(4)+4}{2(4)-1} = 16/7$$

$$\frac{3(5)+4}{2(5)-1} = 19/9$$

$\therefore 7, 10/3, 13/5, 16/7, 19/9$ Ans.

Discrete Assignment 02
Question 10 (Part b):-

(i) $-15, -22, -29, -36 \dots \dots$ 11th term

Sol:-

$$\begin{aligned} d &= -7 \\ \rightarrow a + (n-1)d \\ &= -15 + (11-1)(-7) \\ &= -15 + (10)(-7) \\ &= -15 + (-70) \\ &= -85 \text{ Ans.} \end{aligned}$$

(ii) $a-42b, a-39b, a-36b, a-33b \dots \dots$ 15th term

Sol:-

$$\begin{array}{r} d = -42b \\ a - 39b \\ - \quad \quad \quad \\ \quad \quad \quad + 3b \end{array}$$

$$\begin{aligned} d &= 3b \\ a + (n-1)d \\ &= (a-42b) + (14)(3b) \\ &= a - 42b + 42b \\ &= a \text{ Ans.} \end{aligned}$$

(iii) $4, 3, \frac{9}{4}, \dots \dots$ 17th term

Sol:-

$$\begin{aligned} r &= \frac{3}{4} \\ &= ar^{n-1} \Rightarrow 4\left(\frac{3}{4}\right)^{16} \Rightarrow \frac{3^{16}}{4^{15}} \text{ Ans} \end{aligned}$$

(iv) $32, 16, 8 \dots \dots$ 9th term

Sol:-

$$\frac{16}{32} = \frac{1}{2}$$

$$r = \frac{1}{2} \Rightarrow 32\left(\frac{1}{2}\right)^8$$

$$\begin{array}{r} 8 \times 16 \times 32 \times 1 \\ \hline 256 \\ \hline 128 \\ \hline 64 \\ \hline 32 \end{array} = \frac{8}{32} \text{ Ans}$$

Discrete Assignment or

(6)

Yash Raj

no:- 24K-0737

Question #11 (Part a):-

(i) $T_3 = 10$, $T_5 = \frac{5}{2}$

for GP: $T_n = ar^{n-1}$

Soln:-
 $ar^2 = 10$ $ar^4 = \frac{5}{2}$

$r^2 = \frac{ar^4}{ar^2} = \frac{\frac{5}{2}}{10} = \frac{1}{4}$

$r = \pm \frac{1}{2}$

$a = \frac{10}{r^2} = 10 \times 4 = 40$

$a = 40, r = \pm \frac{1}{2}$ Ans

(part b):-

(i) $T_4 = 7$ & $T_{10} = 31$

Soln:-

$a + 3d = 7$, $a + 9d = 31$

$a + 9d = 31$
 $a + 3d = 7$

 $6d = 24$

$d = 4$

$a + 3(4) = 7$

$a = -5$ Ans

(ii) $T_5 = 8$, $T_8 = \frac{14}{27}$

Soln:-

$ar^4 = 8$

$ar^7 = \frac{14}{27}$

$r^3 = \frac{\frac{14}{27}}{8} = \frac{14}{216}$

$r = \frac{1}{3}$

$a = \frac{8}{r^4} = \frac{8}{(\frac{1}{3})^4} = \frac{8}{\frac{1}{81}} = 648$ Ans

(ii) $T_5 = 86$

$T_{10} = 146$

Soln:-

$a + 4d = 86$

$a + 9d = 146$

$a + 9d = 146$
 $a + 4d = 86$

 $5d = 60$

$d = 12$

$a + 4(12) = 86$

$a = 86 - 48$

$a = 38$ Ans

Question #12:-

(a) Numbers between 286 & 789 divisible by 7.

- Soln.
- Smallest multiple of 7 greater than 286 is $7 \times 37 = 259$
 - Largest multiple of 7 less than 789 is $7 \times 112 = 784$

$112 - 37 + 1 = 76$ numbers.

Sum:-

$$S_n = 7 \times (37 + 38 + \dots + 112)$$

$$S_n = \frac{n}{2} [\text{last term} + \text{first term}]$$

$$S_n = \frac{76}{2} [37 + 112]$$

$$S_n = 8662$$

$$7 \times 8662 = 39634 \text{ Ans.}$$

part b:-

Soln.

$$\frac{1}{n} \rightarrow \text{first term} \quad \frac{n^2 - n + 1}{n} \rightarrow \text{last term}$$

$$S_n = \frac{n}{2} \left[\frac{1}{n} + \frac{n^2 - n + 1}{n} \right]$$

$$\frac{n}{2} \left[\frac{n^2 - n + 2}{n} \right]$$

$$\frac{n^3 - n^2 + 2n}{2n}$$

$$\frac{n(n^2 - n + 2)}{2n}$$

$$= \frac{n^2 - n + 2}{2} \text{ Ans}$$

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Question # 13:-

(a) $\sum_{j=1}^{100} \frac{1}{j}$ Ans.

part (b) (i) $\sum_{k=4}^8 (-1)^k$
80h.

$1 + (-1) + 1 + (-1) + 1 = 1$ Ans

(ii) $\sum_{j=1}^5 j^2$
80h.

$(1)^2 + (2)^2 + (3)^2 + (4)^2 + (5)^2$
 $= 55$ Ans.

Question # 14:-

(a) Ball drop distance:-

$a = 40$, $d = -7$.

~~Ans~~
 $T_5: a + (5-1)d$
 $40 + (4)(-7)$
 $40 - 28$
 $= 12$ feet

$SS = \frac{5}{2} [2a + 4d]$

$= \frac{5}{2} [80 - 28]$

$= \frac{5}{2} \times 52 = 130$
feet
Ans

(b) Population growth:-

$P_0 = 70000$

Annual growth rate = 7%.

After 6 years, as

$P = 70000 \times (1.07)^6 \approx 105051$

Ans

Question # 15:-

(i) $a_n = -2a_{n-1}$, $a_0 = -1$

80h

$a_1 = -2a_0$

$a_1 = -2(-1) = 2$

$a_2 = -2a_1 = (-2)(2) = -4$

$a_3 = -2a_2 = (-2)(-4) = 8$

$a_4 = -2a_3 = (-2)(8) = -16$

Ans

$$(ii) a_n = a_{n-1} - a_{n-2}$$

$$a_0 = 2, a_1 = -1$$

Soln..

$$a_2 = a_1 - a_0$$

$$= -1 - 2$$

$$\boxed{a_2 = -3}$$

$$a_3 = a_2 + a_1$$

$$= -3 + (-1)$$

$$\boxed{a_3 = -4} \text{ Ans.}$$

$$(iii) a_n = 3a_{n-1}^2, a_0 = 1$$

Soln

$$a_0 = 1, a_1 = 3, a_2 = 27$$

$$a_3 = 2187, a_4 = 14348907$$

Ans

$$(iv) a_n = n a_{n-1} + a_{n-2}^2, a_0 = 1, a_1 = 0$$

Soln..

$$a_0 = 1, a_1 = 0, a_2 = 1, a_3 = 3$$

$$a_4 = 13 \text{ Ans.}$$

Part (b):

$$(i) a_n = (-2)^n$$

Soln

$$a_0 = 1, a_1 = -2, a_2 = 4, a_3 = -8$$

Ans.

$$(ii) a_n = 3$$

Soln..

$$a_0 = 3, a_1 = 3, a_2 = 3 \text{ Ans}$$

$$(iii) a_n = 7 + 4^n$$

Soln..

$$a_0 = 8, a_1 = 11, a_2 = 23, a_3 = 47 \text{ Ans}$$

$$(iv) a_n = 2^n + (-2)^n$$

Soln

$$a_0 = 2, a_1 = 0, a_2 = 8, a_3 = 0$$

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Question #16:
(i) . . .

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Question #16:-

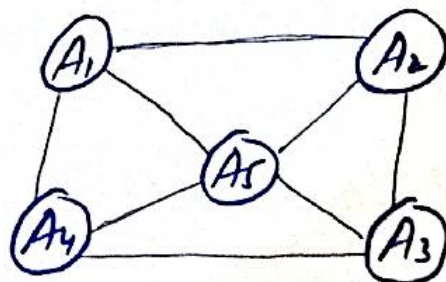
- (i) • Graph has undirected edges
 • Graph has multiple edges
 • Graph has no loops
 • Hence, it is an undirected graph.
- (ii) • Undirected edges
 • No multiple edges
 • No loops
 • Undirected simple graph.
- (iii) • Undirected edges
 • multiple edges
 • 3 loops
 • Undirected Pseudograph
- (iv) • directed graph
 • multiple edges
 • 2 loops
 • directed multi-graph.

Question #17:-

$$A_1 = \{0, 2, 4, 6, 8\} \quad A_2 = \{0, 1, 2, 3, 4\} \quad A_3 = \{1, 3, 5, 7, 9\}$$

$$A_4 = \{5, 6, 7, 8, 9\} \quad A_5 = \{0, 1, 8, 9\}$$

Soln:-



Ans.

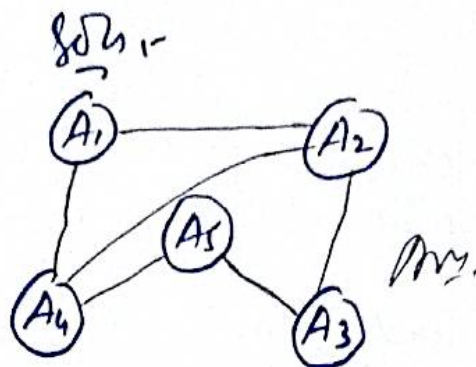
$$(ii) A_1 = \{ \dots, -4, -3, -2, -1, 0 \}$$

$$A_2 = \{ \dots, -2, -1, 0, 1, 2, \dots \}$$

$$A_3 = \{ \dots, -6, -4, -2, 0, 2, 4, 6 \}$$

$$A_4 = \{ \dots, -5, -3, \dots \}$$

$$A_5 = \{ \dots, -6, -3, 0, \dots \}$$



Question #18:-

(a) $\Rightarrow$ (i) Vertices = 5, edges = 13

degree of vertices = $d(a) = 6, d(b) = 6, d(c) = 6, d(d) = 5, d(e) = 3$

Neighbourhood vertices = $N(a) = \{a, b, c\}, N(b) = \{a, c, d, e\}$
 $N(c) = \{b, c, d\}, N(d) = \{b, c, e\}, N(e) = \{a, b, d\}$ Ans.

(ii) Vertices = 9, edges = 12

degree of vertices = $d(a) = 3, d(b) = 2, d(c) = 4, d(d) = 0, d(e) = 6, d(f) = 0$

Neighbourhood vertices = $N(a) = \{c, e, i\}, N(b) = \{c, h\}, N(c) = \{a, e, g, i\},$
 $N(d) = \emptyset, N(e) = \{a, b, c, g\}, N(f) = \emptyset, N(g) = \{c, e\}, N(h) = \{b, i\},$
 $N(i) = \{a, c, h\}$ Ans.

(b)

(i) Indegree:

$d^-(a) = 6, d^-(b) = 1, d^-(c) = 2, d^-(d) = 4, d^-(e) = 0$

outdegree:

$d^+(a) = 1, d^+(b) = 5, d^+(c) = 5, d^+(d) = 2, d^+(e) = 0$ Ans.

(ii) Indegree:

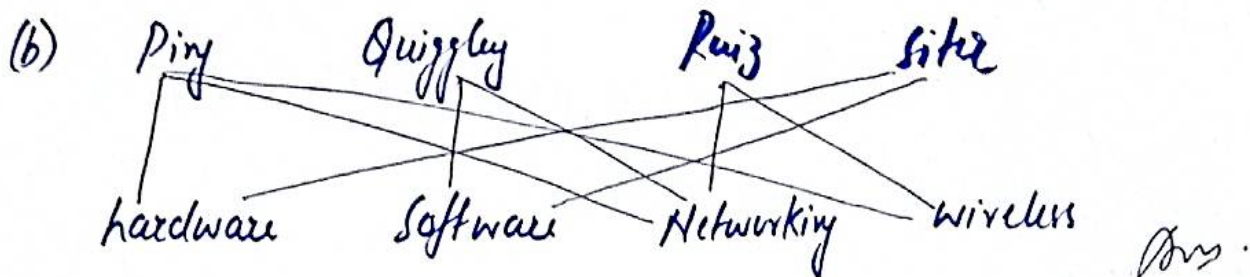
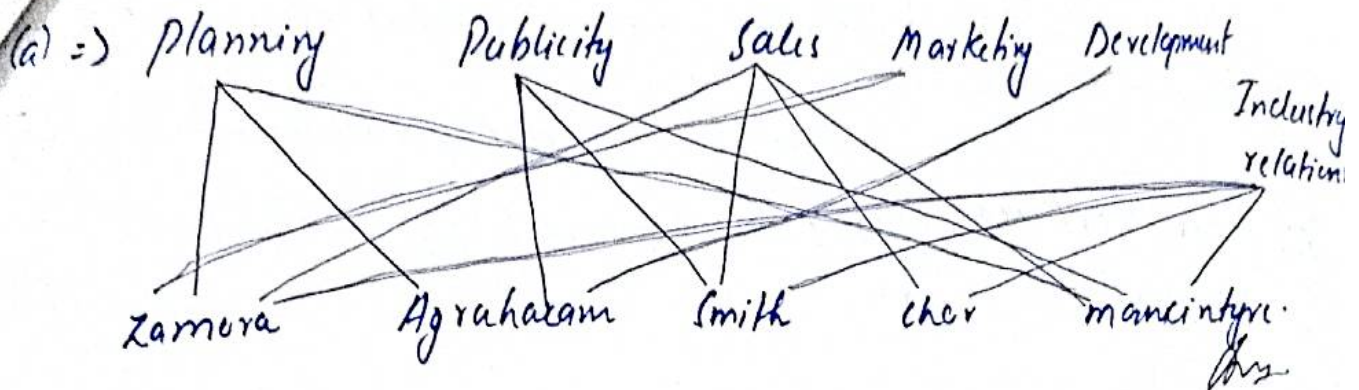
$d^-(a) = 2, d^-(b) = 3, d^-(c) = 2, d^-(d) = 1$

Out-degree:

$d^+(a) = 2, d^+(b) = 4, d^+(c) = 1, d^+(d) = 1$ Ans.

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Question #20:-

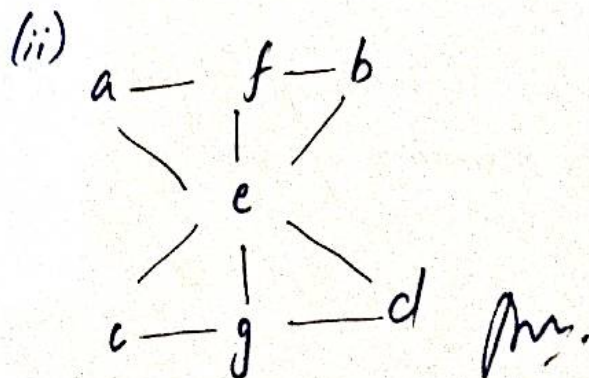
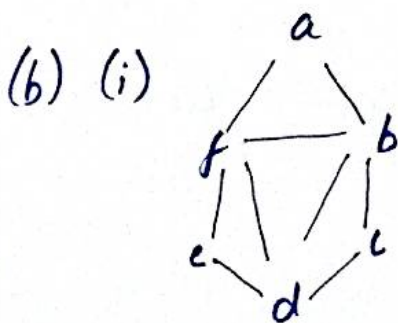
- (i) Not a bipartite graph
- (ii) Bipartite graph
- (iii) Not bipartite
- (iv) Not bipartite

Question 21:-

(a)

- (i) Graph G is Bipartite $\Rightarrow \{u, w, z\}, \{v, x, t\}$
- (ii) Graph H is Bipartite $\Rightarrow \{a, c, e\}, \{b, d, f\}$
- (iii) Graph A is Not a bipartite
 $\Rightarrow \{u, w, y, t\}, \{v, x, z\} \rightarrow$ if i try to color
I will end in doing same color.

(iv) Graph B is Bipartite $\Rightarrow \{d\}, \{e, f, g, h, i, j, k, t\}$ Ans.



Discrete Assignment 02

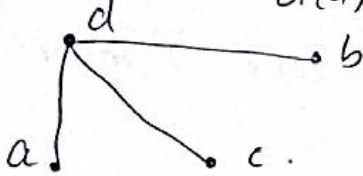
(i) four-vertices: $d(a)=1$, $d(b)=1$, $d(c)=2$, $d(d)=3$.

Soln.
Total degree of graphs $1+1+2+3=7$.

Acc: to handshaking theorem, total degree of graph should be even.

hence graph not possible.

(ii) four vertices: $d(a)=1$, $d(b)=1$, $d(c)=3$, $d(d)=3$.



(iii)

A simple graph with 4 vertices $d(a)=1$, $d(b)=1$, $d(c)=3$, $d(d)=3$ doesn't exist.

Question #23:

(i) No sum of degree must be even but $15 \times 3 = 45$ which is odd. Ans.

(ii) Yes, sum of degree is $4 \times 3 = 12$ Ans.

(iii) people are vertices & handshakes are edges. each person shakes with 9 so, total handshakes.

Total-no: sum of degree. edges $= 90/2 = 45$ Ans.

(iv) 5 people with 2 degree & 10 degree sum.
so edges $10/2 = 5$

for 3 people & possible $4/2$ sum of degree $= 3 \times 2 = 6$ Ans.

(v) edges 10 so, sum of degree $= 10 \times 2 = 20$ & each degree has 4 so number of vertices $= 20/4 = 5$ Ans.

Yash Ray

no:- 24k-0737.

Question #24 :-

(a) $W \rightarrow$ Connected
 $X \rightarrow$ Not Connected
 $Y \rightarrow$ Connected
 $Z \rightarrow$ Connected Ans

(b) $W \rightarrow$ Cycle-free (No closed loop)
 $X \rightarrow$ has cycles inside graph and
 $Y \rightarrow$ has a triangle ABC
 $Z \rightarrow$ has cycles.

(c) $\rightarrow$ cycle free graph $\Rightarrow$ w-acy Ans
 loop free - w-acy
 has loop : Z acy.

Ans

(d) No loops
 No parallel edges
 No multiple edges Ans
 Undirected

Question #25 :-

- (i) Vertices = 5, edges = 7, degree sequence = $\{4, 3, 3, 2, 2\}$ conditions are satisfied

hence isomorphic.

$$g: v(l_i) \rightarrow v(l'_i) \Rightarrow g(v_1) = w_2, g(v_2) = w_3, g(v_3) = w_2, g(v_4) = w_1, g(v_5) = w_4$$

- (ii) Vertices = 6, edges = 7, degree sequence = $\{3, 3, 3, 2, 2, 1\}$ conditions satisfied.

hence isomorphic.

$$g: v(l_i) \rightarrow v(l'_i) \Rightarrow g(v_1) = u_5, g(v_2) = u_2, g(v_3) = u_4, g(v_4) = u_1, g(v_5) = u_6$$

- (iii) Vertices = 7, edges = 9, degree sequence = $\{4, 3, 3, 2, 2, 2, 2\}$ conditions satisfied.

hence isomorphic.

$$g: v(l_i) \rightarrow v(l'_i) \Rightarrow g(v_1) = u_5, g(v_2) = u_4, g(v_3) = u_3, g(v_4) = u_2, g(v_5) = u_7, g(v_6) = u_1, g(v_7) = u_6$$

- (iv) Vertices = 5, edges = 7, degree sequence = different.

hence not isomorphic

Ans

Question #48 :-

$$f(a) = 2a + 3$$

$$g(a) = 3a + 2$$

(a) $f \circ g(a) = 6a + 7$ $g \circ f(a) = 6a + 11$

(b) Both are linear, injective not surjective.

(c) They are not invertible.

Ans

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Question #28:

(a) ABCDA: 125
ABDCA: 140
ACDBA: 140
ADBCA: 155
So min is ABCDA: 125

(b) ABCDA: 108
ADBCA: 141
ADLBA: 119
ACBDA: 141
ABLDA: 97

shortest

(c) ABCDA: 95
ABDLA: 80
ADBCA: 90
ADLBA: 95

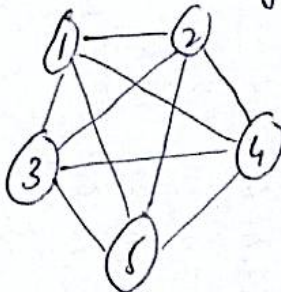
shortest

Question #29:

part (a):- This is euler path & A(source) & (Destination) E have odd degree & every other have even. so, yes it is possible.

part (b):- Yes, possible because A & b have odd edges.

part (c):- Every degree has even degree it Euler-circuit.



2 → 5 → 1 → 4 → 3 → 2

Question #26:

A ⇒ E: 5, V: 5, Sequence: 2, 3, 3, 2, 2
F ⇒ E: 4, V: 5, Sequence: 1, 2, 3, 1, 1
K ⇒ E: 4, V: 5, Sequence: 1, 1, 4, 1, 1
M ⇒ E: 4, V: 5, Sequence: 2, 2, 2, 1, 1
R ⇒ E: 5, V: 5, Sequence: 3, 3, 1, 1, 1

S ⇒ E: 4, V: 5, Sequence: 1, 2, 2, 2, 1
T ⇒ E: 4, V: 5, Sequence: 3, 1, 1, 2, 1
V ⇒ E: 4, V: 5, Sequence: 2, 2, 2, 1, 1
X ⇒ E: 4, V: 5, Sequence: 4, 1, 1, 1, 1
Z ⇒ E: 4, V: 5, Sequence: 1, 2, 2, 2, 1

X & X are isomorphic, S, M, V, Z are isomorphic, F & T are isomorphic

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
a	2,a	2,a	4,a	1,a	inf	inf	inf	inf	inf	inf	inf	inf	inf	inf	inf	inf	inf	inf	inf	inf	inf	inf	inf	inf	inf	
adb	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbe	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbee	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbyg	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbeqf	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbeqfh	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbeqfhk	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbeqfhkl	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbeqfhkkl	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbeqfhkkliv	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbeqfhkklivj	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbeqfhkklivjm	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbeqfhkklivjmn	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbeqfhkklivjmnp	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbeqfhkklivjmnpq	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbeqfhkklivjmnpqo	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbeqfhkklivjmnpqot	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbeqfhkklivjmnpqotk	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	
acbeqfhkklivjmnpqotkz	2,a	2,a	4,a	1,a	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	3,b	

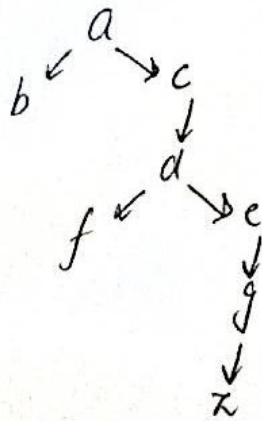
2027

Name: Yash Raj

Roll-no: 241C-0737.

Question 27:- (a) Shortest path = $2+1+3+8+2+2+2=20$ Ans.

(b)	D(b)	D(c)	D(d)	D(e)	D(f)	D(g)	D(z)
a	4, a	3, a					
ac	4, a		6, c	9, c			
acb			6, c	9, c			
acbd				7, d	11, d		
acbde					11, d	12, e	
acbdef						12, e	8, f
acbdefg							16, f
acbdefgz							



path a to z:
 a-c-d-e-g-z
 length = $3+3+1+1+4=16$
 Ans.

part(c):-

a

ac

acb

acbf

acbf c

acbf e z

acbf e z d

D(b)

12, a

D(c)

10, a

D(f)

12, a

D(e)

13, c

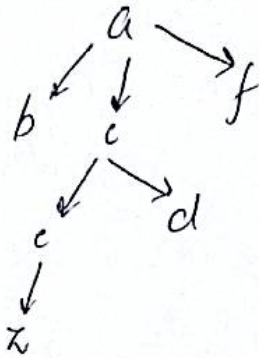
D(f)

21, c

D(z)

21, f

21, e



path = a-c-e-z

length = 10 + 3 + 6 = 19

Question #31 :-

- (a)
- (i) It's a Euler circuit as every vertex has even degree.
 - (ii) Not a Euler circuit because some vertex has odd degree.
 - (iii) It's a circuit (Euler).

- (b)
- (i) Path doesn't exist as u & w have odd degree along with all vertices.
 - (ii) Yes it is a Euler-path because only u & w have odd degree.

Question #30 :-

- (a) (i) $v_1 - v_2 - v_3 - v_7 - v_4 - v_5 - v_6 - v_2 - v_1$ (H-circuit).
 $v_1 - v_8 - v_6 - v_4 - v_7 - v_5 - v_2 - v_3$ (H-Path)

(ii) $b - a - c - h - g - f - d$ (H-Path) $\rightarrow$ No Circuit.

(iii) $a - b - c - d - e - f - g - a$ (H-circuit).
 $a - b - c - d - g - f - c$ (H-path).

- (b) (i) $a - h - g - f - e - d - c - b - a$ (H-circuit) $\rightarrow$ (H-Path) $\rightarrow a - b - h - g - f - e - d - c$.

(ii) No Circuit. $d - e - f - e - j - c - b - h - a$ (H-path).

(iii) H-circuit = $a - x - b - y - c - z - a$

H-path = $a - x - b - y - c - z$

Ans

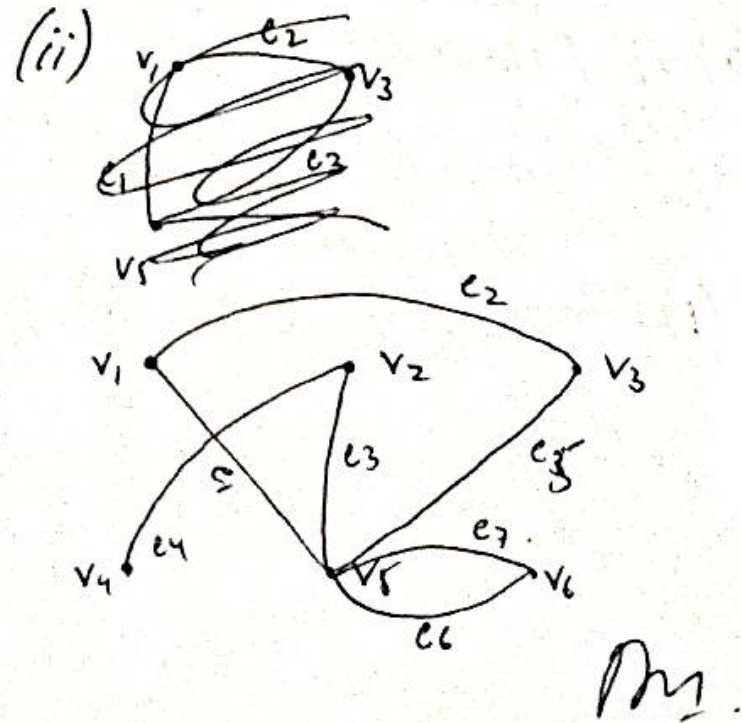
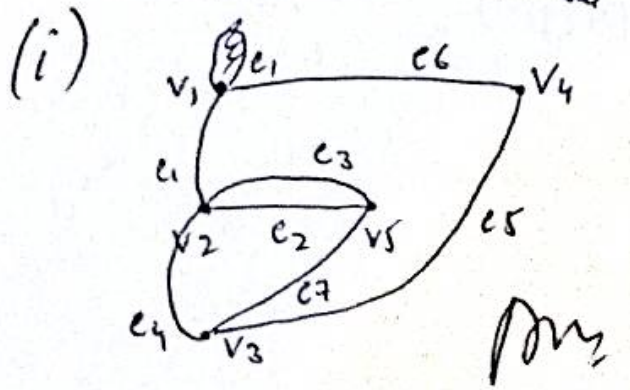
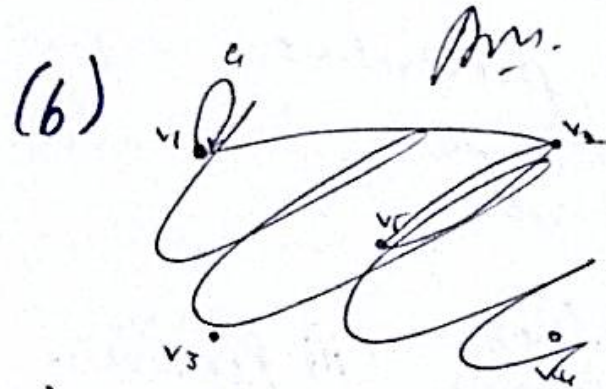
Question #32:-

(a) (i)

	e_1	e_2	e_3	e_4	e_5	e_6	e_7
v_1	1	1	1	0	0	0	0
v_2	0	0	0	0	1	1	1
v_3	0	1	1	1	0	0	0
v_4	0	0	0	1	1	0	0
v_5	0	0	0	0	0	1	0
v_6	1	0	0	0	0	0	1

(ii)

	e_1	e_2	e_3	e_4	e_5	e_6	e_7	e_8
v_1	1	1	1	0	0	0	0	0
v_2	0	1	1	1	0	1	1	0
v_3	0	0	0	1	1	0	0	0
v_4	0	0	0	0	0	0	1	1
v_5	0	0	0	0	1	1	0	0



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Question #33:- (part a):-

(i)

	a	b	c	d
a	1	1	1	1
b	0	0	0	1
c	1	1	0	0
d	0	1	1	1

Ans

List:- $a \rightarrow a, b, c, d$ $b \rightarrow d$ $c \rightarrow a, b$ $d \rightarrow b, c, d$ Ans

(iii)

	a	b	c	d
a	0	3	0	1
b	3	0	1	0
c	0	1	0	3
d	1	0	3	0

 $a \rightarrow b, b, b, d$ $b \rightarrow a, a, a, c$ $c \rightarrow b, d, d, d$ $d \rightarrow a, c, c, c$ Ans

(ii)

	a	b	c	d	e
a	0	1	0	1	0
b	1	0	1	1	1
c	0	1	1	0	0
d	1	0	0	0	1
e	0	0	1	0	1

Ans.

List:- $a \rightarrow b, d$ $b \rightarrow a, c, d, e$ $c \rightarrow b, c$ $d \rightarrow a, e$ $e \rightarrow c, c$ Ans

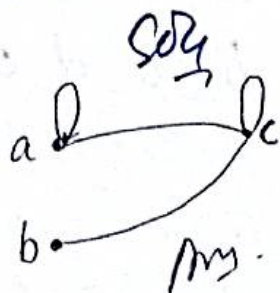
(iv)

	a	b	c	d
a	1	0	2	1
b	0	1	1	2
c	2	1	1	0
d	1	2	0	1

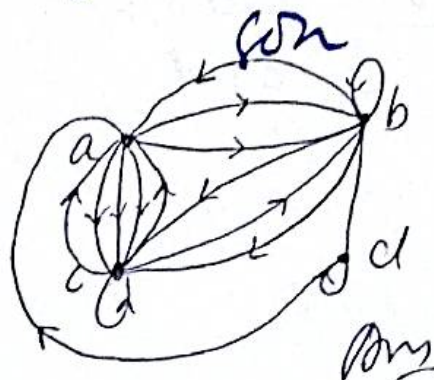
 $a \rightarrow a, c, c$ $b \rightarrow b, c, d, d$ $c \rightarrow a, a, b, c$ $d \rightarrow a, b, b, d$ Ans

Part (b) :-

$$(i) \begin{matrix} a & b & c \\ a & 1 & 0 & 1 \\ b & 0 & 0 & 1 \\ c & 1 & 1 & 1 \end{matrix}$$



$$(ii) \begin{matrix} a & b & c & d \\ a & 0 & 2 & 3 & 0 \\ b & 1 & 2 & 2 & 1 \\ c & 2 & 1 & 1 & 0 \\ d & 1 & 0 & 0 & 2 \end{matrix}$$



Question #34 :-

(a) (i) Complete bipartite graph is never planar & $e \leq 3v-6$
 $\Rightarrow 6 \leq 15-6$
 $\Rightarrow 6 \leq 9$

(ii) it has no circuit of length 3 so $e \leq 3v-4$.
 $6 \leq 3(6)-4$
 $6 \leq 18-4$
 $6 \leq 14$
 So, not a planar.

(b) $G \Rightarrow v=4$, edges $=6$, degree-sequence $=3,3,3,3 \rightarrow$ odd degree
 $R=4$ $R = E-v+2 \Rightarrow 4 = 6-4+2$ $\boxed{4=4}$ Euler not applicable.

$H \Rightarrow v=6$, $E=9$ $R=5$ $R = E-v+2 \Rightarrow 5=5$

$J \Rightarrow v=5$ $e=10$ $R=7$.

$R = E-v+2$
 $\boxed{7=7}$ pm .

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Question #35:-

- (a) 3 (b) 0 (c) 5 (d) $u \in v$ (e) d (f) $k \in L$
 (g) m, s, t, x, y (h) $\begin{matrix} \text{a, b, c, d, e, f, h, i} \\ \text{a, d, h, n, v, k, m, n, o, t, u} \end{matrix}$ (i) j, l, p, q, r, s, u, w, x, y, z

Question #36:-

- (a) (i) $v_0 \rightarrow v_5 \rightarrow v_6 \rightarrow v_7 \rightarrow v_2$ total = 61 Pms
 $\quad \quad \quad \downarrow$
 $\quad \quad \quad v_4 \rightarrow v_3 \rightarrow v_1$
 (ii) $v_0 \rightarrow v_1 \rightarrow v_3 \rightarrow v_2 \rightarrow v_8 \rightarrow v_7 \rightarrow v_6 \rightarrow v_3 \rightarrow v_9 \rightarrow v_5$

(b) (i) d, f = 1

a, c = 2
 a, b = 3
 b, c = 3 x
 c, d = 4
 d, a = 5
 c, a = 6
 c, f = 6 x
 a $\rightarrow$ b
 $\quad \downarrow$
 $\quad c \rightarrow d \rightarrow e$
 $\quad \quad \downarrow$
 $\quad \quad f$
 lengths
 $1 + 2 + 3 + 4 + 5$
 $= 15$ Pms.

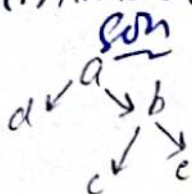
(ii) g, f = 1

a, b = 3
 a, c = 3
 b, c = 4 x
 a, g = 4
 c, f = 4 x
 g, c = 5 x
 h, c = 7
 c, e = 7
 c, d = 10
 d, e = 11 x
 d, f = 12 x
 g $\rightarrow$ f $\rightarrow$ c $\rightarrow$ a $\rightarrow$ b $\rightarrow$ c $\rightarrow$ d
 $\quad \quad \quad \downarrow$
 $\quad \quad \quad 1 + 3 + 3 + 4 + 7 + 10$
 $\quad \quad \quad = 28$
 Pms

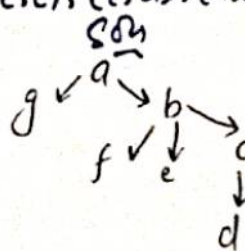
Question #38:-

- (a) (i) pre-order = a, b, c, k, l, m, f, g, h, r, s, c, d, (ii) pre-order = a, b, d, e, i, j, m, n, o, f, h, k, p, g, l
 (ii) Inorder = k, e, l, m, b, f, r, n, s, g, a, c, o, h, d, i, p, j, r, n, e, d, b, i, e, m, j, n, o, a, f, c, k, h, g, l
 (iii) Post-order = k, l, m, e, f, r, s, n, g, h, c, a, l, i, Post-order = d, i, m, n, o, j, e, b, f, g, p, i, c, a

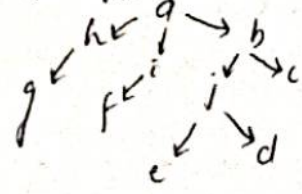
(b) (i) Remove (d, b) & (c, e)



(ii) Remove (g, b), (g, f), (f, e), (c, e), (e, d), (a, d), (g, d)

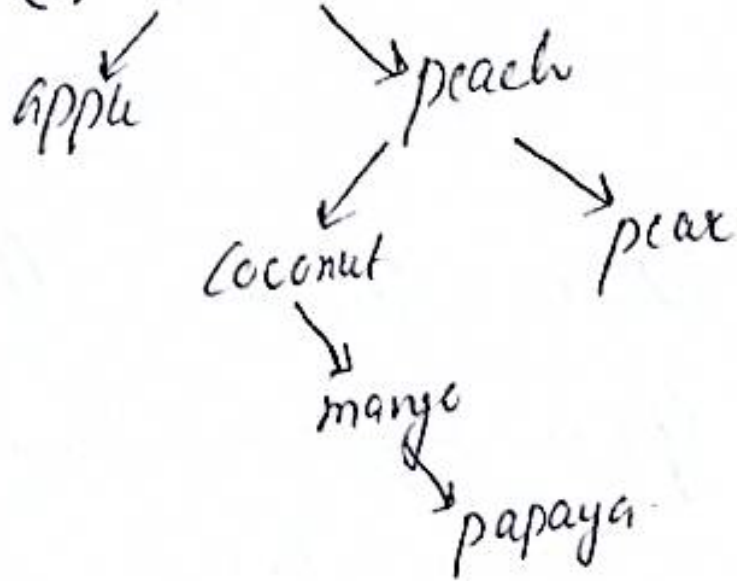


(iii) Remove:-

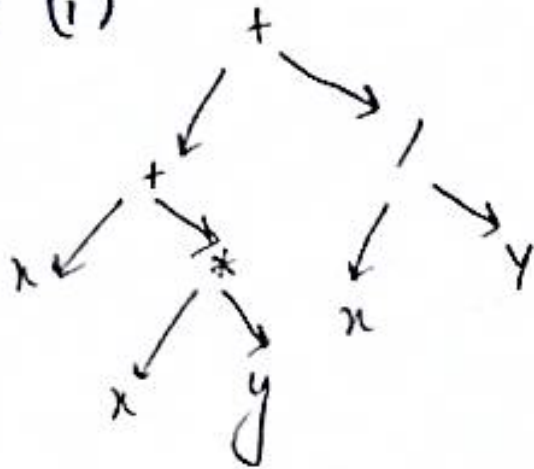


Question #37:-

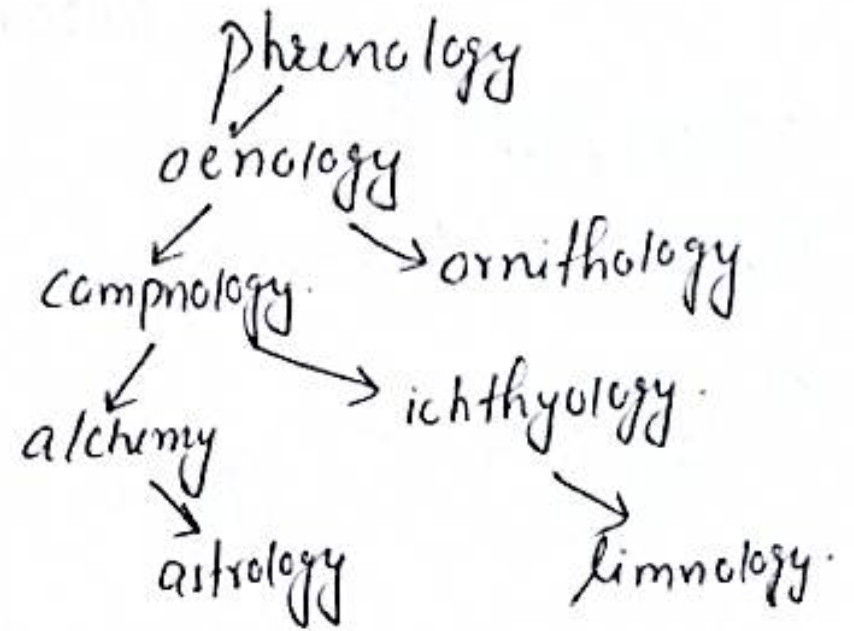
(a) (i) banana



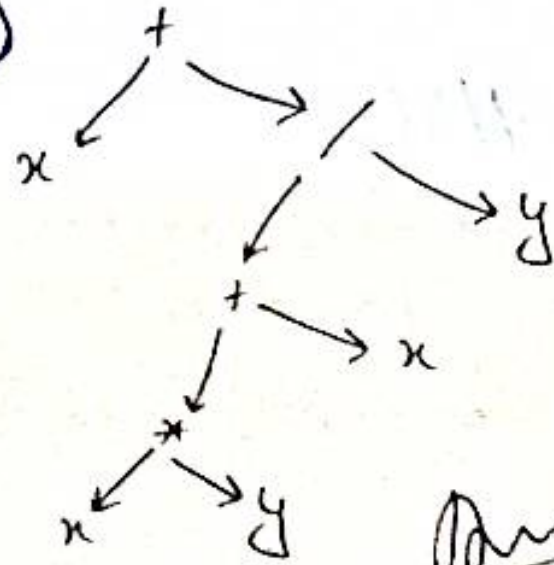
(b) (i)



(ii)



(ii)



Am.

Question #40 :-

(a)

(i) $(n + ny) + (n/y)$

prefix $++n * ny / ny$

postfix $ny * n / y$

(ii) $n + ((ny + n) / y)$

prefix $++n / + * ny ny$

postfix $ny * n / y$ Ans

(b) (i) prefix $: + - 98 / 625 + 13 = 4$

(ii) postfix $: 121 * 14 * / 5 124 / 3$ Ans

Question #39 :-

(a) Edges $: vertices - 1 \Rightarrow 10000 - 1 = 9999$

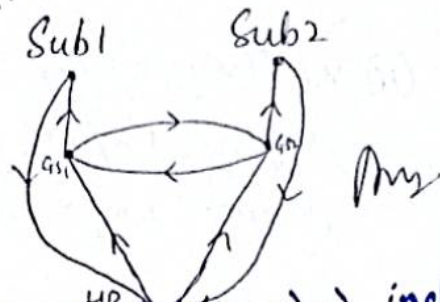
(b) Edges $: d_i = 2(10000) = 20000$ edges

(c) Total vertices $: m \times i + 1 \Rightarrow 5(100) + 1 = 501$

(d) $n = m + 1 \Rightarrow 1 = 100 - 33 = 67$ Ans

Question #43:-

(a)



- (b) GS1 (German Soldier 1) $\Rightarrow$ indegree = 2
 GS2 (= = 2) $\Rightarrow$ indegree = 2
 HP (Head of Platoon) $\Rightarrow$ indegree = 2
 Sub1 (Soldier 1) $\Rightarrow$ indegree = 1
 Sub2 (Soldier 2) $\Rightarrow$ indegree = 1

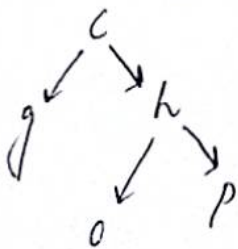
out-degree = 2
 out-degree = 2
 out-degree = 2
 out-degree = 1
 out-degree = 1

- (c) Neither it equivalence nor partial order because for both we need reflexive property to be true.

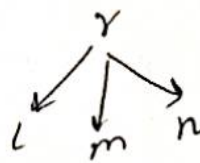
Question #41:-

- (a) No it's not because it doesn't follow the condition that the parents should have 0 or m child.
 (b) No because it has leaves on level 1 to level 4.
 height = 5 so acc: BBT we should only have leaves on 5th or 4th level.

(c) (i) subtree of c



(ii) subtree of f



(iii) subtree of e



Question #46:-

(a)

1) no. of vertices = 8, number of edges = 12

2) Sequence of degrees = 3, 3, 3, 3, 3, 3, 3, 3

3) Mapping:-

$$r(A) = 1$$

$$r(B) = 2$$

$$r(C) = 3$$

$$r(D) = 4$$

$$r(W) = 5$$

$$r(X) = 6$$

$$r(Y) = 7$$

$$r(Z) = 8$$

} This proves that G and H are isomorphic.

(b)

(i) No it is not a Complete Bipartite graph because each vertex or disjoint set $U = \{A, B, C, D\}$ is not connected to all vertex or distinct disjoint set.

(ii) It's not Euler circuit because vertex degree should be even but here all are odd.

Yes, hamilton circuit exists:-

$1 \rightarrow 2 \rightarrow 4 \rightarrow 7 \rightarrow 6 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow 1$ Ans.

Question #44:-

(a)

N'	A	B	C	D	E	F	S
H	(3, H)	(2, H)	(6, H)	(3, B)	(8, B)	-	-
HB				(3, B)	(8, B)	(7, D)	
HRD					(8, B)	(7, D)	-
HRDA					(7, C)	(7, D)	-
HRDAC							(11, E)
HRDAEC							(9, F)
HRDAECF							

(b) $deg(H) = 3$ $deg(A) = 2$ $deg(B) = 3$
 $deg(C) = 2$ $deg(D) = 3$ $deg(E) = 4$
 $deg(F) = 3$ $deg(S) = 2$

→ odd-degrees vertex (H, B, D, F).

→ As it doesn't have even degree so it is not euler.

(c) By visiting each vertex we can't reach to the source or to any other vertex like home to school crossing every vertex. so it's neither hamilton circuit nor path.

1) Shortest path is 9 from H vertex
 ans: (9, f) Ans.

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Question #46:-

$$A = \{a, b, c, d\} \quad B = \{a, b, c, d\}$$

$$(a) f(a)=b, f(b)=a, f(c)=c, f(d)=d$$

Sol:-

- (i) Range $\{a, b, c, d\} \Rightarrow \{ \text{every element of } B \text{ appears as an image} \}$
 (ii) Injective: Yes
 Surjective: Yes (Range: co-domain)
 Bijective: Yes
 (iii) Inverse exists (self-inverse) $\Rightarrow f^{-1}(a)=b, f^{-1}(b)=a, f^{-1}(c)=c, f^{-1}(d)=d$ Ans.

$$(b) f(a)=b, f(b)=b, f(c)=d, f(d)=c$$

Sol:-

- (i) Range $\{b, c, d\}$
 (ii) Injective: No
 Surjective: No
 Bijective: No
 (iii) Inverse not possible. Ans.

$$(c) f(a)=d, f(b)=b, f(c)=c, f(d)=d$$

Sol:-

- (i) Range $\{a, b, c, d\}$
 (ii) Injective: No
 Surjective: No
 Bijective: No
 (iii) No - Inverse. Ans.

$$(d) f(a)=c, f(b)=a, f(c)=b, f(d)=d$$

Sol:-

- (i) Range $\{a, b, c, d\}$
 (ii) Injective: Yes
 (iii) Bijective: Yes
 Surjective: Yes

$$(iv) \text{Inverse exists } \Rightarrow f^{-1}(a)=b, f^{-1}(b)=c, f^{-1}(c)=a, f^{-1}(d)=d$$

Ans.

Question #47:

(a) $f(x) = \lfloor \frac{x^2}{3} \rfloor$, find $f(s)$ if:

(i) $S = \{-2, -1, 0, 1, 2, 3\}$

$\lfloor \frac{(-2)^2}{3} \rfloor = \lfloor \frac{4}{3} \rfloor = 1$

$\lfloor \frac{(-1)^2}{3} \rfloor = \lfloor \frac{1}{3} \rfloor = 0$

$\lfloor \frac{0^2}{3} \rfloor = 0$ $\lfloor \frac{(2)^2}{3} \rfloor = 1$

$\lfloor \frac{1^2}{3} \rfloor = 0$ $\lfloor \frac{(3)^2}{3} \rfloor = 3$
 $f(s) = \{0, 1, 3\}$

(iii) $S = \{1, 5, 7, 11\}$

$\lfloor \frac{1^2}{3} \rfloor = 0$

$\lfloor \frac{25}{3} \rfloor = 8$ $f(s) = \{0, 8, 16, 40\}$

$\lfloor \frac{49}{3} \rfloor = 16$

$\lfloor \frac{121}{3} \rfloor = 40$

(ii) $S = \{0, 1, 2, 3, 4, 5\}$

$\lfloor \frac{(0)^2}{3} \rfloor = 0$

$\lfloor \frac{1^2}{3} \rfloor = 0$

$\lfloor \frac{4}{3} \rfloor = 1$

$\lfloor \frac{8}{3} \rfloor = 2$ $f(s) = \{0, 1, 2, 3, 8\}$

$\lfloor \frac{25}{3} \rfloor = 8$

(iv) $S = \{2, 6, 10, 14\}$

$\lfloor \frac{4}{3} \rfloor = 1$ $\lfloor \frac{36}{3} \rfloor = 12$

$\lfloor \frac{100}{3} \rfloor = 33$ $\lfloor \frac{196}{3} \rfloor = 65$

$f(s) = \{1, 12, 33, 65\}$

(v) $\lfloor \frac{3}{4} \rfloor$

$\lfloor \frac{-7}{8} \rfloor = 0.875$

(vi) $\lfloor -1 \rfloor$

-1

(b) $\lfloor \frac{3}{4} \rfloor$

$\lfloor \frac{3}{4} \rfloor = 0.75$

(ii) $\lfloor \frac{7}{8} \rfloor$

$\lfloor \frac{7}{8} \rfloor = 0$

(iii) $\lfloor \frac{-3}{4} \rfloor$

$\lfloor \frac{-3}{4} \rfloor = 0$

(vii) $\lfloor \frac{1}{2} + \lfloor \frac{3}{2} \rfloor \rfloor$

$\lfloor 0.5 + 2 \rfloor = \lfloor 2.5 \rfloor = 2$

(viii) $\lfloor \frac{1}{2} \cdot \lfloor \frac{5}{2} \rfloor \rfloor$

$\lfloor 1 \rfloor = 1$

(c) (i) $\lfloor -n \rfloor = -\lceil n \rceil$

$\lceil n \rceil$ is the smallest integer m such that $n \leq m$.
 if $\lceil n \rceil = n$ then $n-1 < n$
 multiply -1 :
 $-(n-1) > -n > -n$
 $-n < -n - (n-1)$

$\lfloor -n \rfloor = -n = -\lceil n \rceil$

$\lfloor -n \rfloor = -n = -\lceil n \rceil$

$\lfloor -n \rfloor = -n = -\lceil n \rceil$

proved

(ii) $\lceil -n \rceil = -\lfloor n \rfloor$

$\lfloor n \rfloor = m$ means $m \leq n < m+1$

multiply -1 :
 $-m \geq -n > -(m+1)$

$-(m+1) < -n \leq -m$

$\lceil -n \rceil = -m = -\lfloor n \rfloor$ proved

Question #48:

$$f(a) = 2a + 3$$

$$g(a) = 3a + 2$$

- (a) $f \circ g(a) = 6a + 7$ $g \circ f(a) = 6a + 11$
- (b) Both are linear, injective not surjective.
- (c) They are not invertible.

Ans.

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Question #49:(a) Functions:(i) Temperature conversion: $F(n) = \frac{9}{5}n + 32$.

→ used in weather forecasting & engineering.

(ii) Salary calculation:

 $s(n) = \text{base salary} + (\text{bonus-rate} \times n)$.

→ used in payroll systems to compute employee salaries.

(b) Relations:

(i) Social Networks: friends with follows relationship on social media platforms.

(ii) Student Course Enrollment: Relation b/w courses & students they have enrolled.

(c) Sequences:

(i) Bank Account Growth: Monthly balances following a pattern like an arithmetic or geometric sequence when interest is applied.

(ii) Traffic Light Timing: The time intervals for signals.

(d) Series:

(i) Loan or EMI calculation: monthly payments are based on the formula for sum of a geometric series.

(ii) Population Growth models:

Used to predict total population over time.

(f) Trees:(e) Graph Theory:

(i) Google Maps: Roads & intersections are modeled as vertices & edges to find shortest-path.

(ii) Social media Networks: Each user is a Node & friendships are edges - used to analyze connectivity.

(i) File System Structure: folders & files in computers are organized in a tree like hierarchy.

(ii) Decision Trees in AI:

Used in machine learning & data science for making predictions.